Handoffs Decision Optimization of Mobile Celular Networks

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***Abstract*—A neural network (NN) approach is proposed in this paper to optimize the operation of cellular networks. First, we derive an analytical equation to establish the effect of essential handoff contributory factors. Data of these factors were then obtained from base stations of an operational network carrier, and trained on a backpropagation NN algorithm. Results obtained revealed that the network performance improved with big data and increase in number of neurons for SINR data, but, increasing the number of layers degraded the system performance. However, with relative signal strength data, improved performance was achieved for big data and increase in number of layers for normalized and un-normalized data. Finally, self organizing map was explored to visualize the existing system for the purpose of improving further, its performance.**

Keywords—CDMA; handoff optimization; multilayer perceptron, neural network, self organizing map;

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I. INTRODUCTION

The cellular concept has organized mobile telephony differently. Today, most low-power transmitters are massively distributed across coverage areas, as oppose to earlier deployment of powerful transmitters with wider coverage. Dividing a dense subscribers' region into cell pockets with low-power transmitters has contributed to an increase in the network capacity, and consequently, accommodates more users.

The key component of any mobile communication network is mobility. Mobility significantly affects the network protocol, its connectivity, as well as the performance of its mechanism. Hence, position and velocity of users become crucial, as these factors contribute to optimizing the performance of the system control algorithms [1-2]. Soft handoff provides improved communication quality by stabilizing transmission and enhancing effective mobility management through a reduced network signal load and data loss. It allows for the overlapping of the repeater coverage zones, such that every mobile user appears within the coverage range of at least one of the neighboring base stations [3]. This technique guarantees seamless transition from one cell to another and is characteristic of Code Division Multiple Access (CDMA) systems. The seamless continuity of calls largely depends on the handoff decision, and this decision becomes more challenging at the cell boundary, as calls in this domain are most likely to experience degraded signal quality, due to frequent link switching between the mobile users (MUs) and the respective base stations. Given the relative short period the mobile user requires to negotiate a handoff, handoff decisions are primarily determined within the current domain of the MU or the locality of the transient loop. The decisions must be very fast, as late handoffs are not tolerable. As the cell size decreases, the number of handoff calls increases, owing to the continuous mobility of MUs across numerous base stations. Hence, setting a fixed threshold value would certainly yield unsatisfactory results because the thresholds changes with users' mobility. A handoff approach with variable threshold mechanism is required to serve this purpose. Consequently, intelligent techniques are preferred over traditional techniques to dynamically regulate the thresholds as required. Intelligent techniques posses the ability to resolve complex patterns with imprecise and noisy inputs.

A. CDMA Handoff

CDMA systems implement the spread spectrum technique known as the universal frequency reuse, which permits every user to share the available radio channels, thus, making handoff connections relatively permanent and stable, compared to other cellular technologies. The soft handoff solution however presents two new challenges, (i) the Multiple Access Interference (MAI) caused by the simultaneous transmission of all users in the same band and the synchronous received signals with a non-orthogonal random sequence in the uplink, and (ii) the near-far effect – a handoff effect were communication pilot signals are serviced by stronger and farther cells. The first problem can be resolved by the exclusion of identical frequencies from each other, while the second problem can be minimized by a power control mechanism or completely eliminated using an efficient power control mechanism. Several figures of merit have been

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proposed to maintain handoff decisions in cellular networks. The most prominent ones are factors that affect the decision process itself, and include, the Received Signal Strength (RSS), Signal to Interference and Noise Ratio (SINR), cell traffic, velocity of the mobile user, Bit Error Rate (BER), bandwidth coverage, and distance of the mobile user from the base station. Amongst these merits, RSS appears to be the most commonly used metric [4, 5, 6], because estimating the RSSs of nearby base stations is very crucial to reaching a handoff decision.

Integrating intelligence into CDMA systems has been one of the most recent developments and new technique explored to suppress narrow-based and MAI in interference limited systems. In this paper, we explore this new technique to investigate an operational CDMA carrier. We review related works, provide an analytical estimate, present the system model design, and offer extensive simulations of the system under study. With the proposed system's capabilities, mobile operators would effectively deliver higher bandwidth applications with streaming and broadband services, and manage optimally, traffic and handoffs requests.

B. Problems and challenges in developing nations:

The purpose of our design is to ensure that the stochastic nature of the network is explored to reveal the true state of the system and enable real time intervention. As observed, existing network carriers would focus on the infrastructure and profit making, instead of QoS improvements. This is usually the case in developing countries, for instance, where the geography and topology of the area were not well planned before deployment and does not permit such huge investments. Also the population spread is highly skewed to the urban areas, which become denser than others, due to heavy migration of persons from the rural to urban areas due to unbalanced economy. In addition, most urban areas lack good master plans to regulate the type of buildings in certain areas where telecomm equipment are deployed, which leaves the carriers with no other alternative than to plant cell stations close to leaving premises. The outcomes of these practices are severe health hazards and degraded network service.

On the issue of proper regulation by the appropriate authorities, not much has been done to enforce the standards, and compensate affected customers. In our case study (Nigeria), for instance, most customers resort to having more than one phone line, to preempt a failure in network service. On the part of the carriers, it may lead to the setting of unrealistic targets just to portray an efficient network. Optimization therefore remains the only viable option in this circumstance, and is the goal of this paper.

II. RELATED WORKS

Efficient handoff decision is crucial to prevent unnecessary handoff failures in wireless communication networks. Hence, maintaining seamless handoffs while still achieving the expected quality of service (QoS) can be very demanding. Using a random walk model and fuzzy theory, a handoff system which employs three parameters for handoff decision (signal strength from the current base station, signal strength from the neighboring base station and distance between the mobile station and the base station) has been proposed in [5]. Performance evaluation via simulation showed that the proposed system avoided the 'ping-pong' effect and had a good handoff decision mechanism.

Handoff drop (HOD) rate and bandwidth reservation have been identified in [7] as two important metrics for evaluating the handoff performance in cellular systems. The authors introduced an ad hoc mechanism into cellular systems to help minimize HOD rate and incorporated traffic information from neighboring base stations with an algorithm that obtains the minimum bandwidth reservation for each base station. They further utilized the ad hoc network to gather traffic load information of neighboring cells, such that handoff calls effectively select the best paths to the target BSs. A demonstration of their scheme may significantly improve the system performance.

An algorithm for controlling handoff in mobile wireless networks has been described in [8]. The authors discussed the conventional handoff control algorithm and used average values of the received signal strength for handoff decisions, where mobile terminals can take other measurements, such as distance to base stations, which are important for handoff decisions. The algorithm uses a fuzzy logic decision process and considered the simulation of two cases (2 and 4 base stations). Results of their simulations showed that in both cases, the proposed algorithm provided better performance than conventional ones.

An intelligent mobility management scheme for Mobile Wireless InterNet (MWIN), where the two major standards of MWIN, the mobile IP and wireless LAN are employed for the purpose of handoff minimization is presented in [10]. It was observed that both schemes solved problems independently at different layers, thereby creating further problems such as delayed handoff, packet loss, and inefficient routing. These problems were identified and analysis performed on real handoff data within a typical MWIN. A handoff architecture extending the features of both standards' handoff mechanism was also proposed. This architecture consists of mobile IP extensions and a modified wireless LAN handoff algorithm that provides a link between the various layers to prevent packet losses and reduce handoff latency.

A fuzzy logic based scheme for selecting the target base station during handoff is presented in [9]. The scheme considered three criteria, namely, received power levels, user population and used bandwidth of each base station to implement a fuzzy handoff decision. They also compared the conventional power level based handoff and user population based handoff schemes. Simulation results showed that the fuzzy logic based scheme resulted in uniform user load at the base stations and optimized the number of base station switching.

The artificial neural networks (ANNs) have emerged as fast computation tools with learning and adaptive capabilities. They are preferred over fuzzy logic because of their ability to deal with noisy data. Research works exploring neural networks (NNs) for handoff optimization have not been deeply researched. The introduction of NNs to call handoff management was attempted in [6]. The authors used an

Artificial Neural Network (ANN) to ensure fast and accurate handoff decision, but concentrated only on the received signal strength data. Their results – obtained using a backpropagation neural network model, indicated optimum results. In [11], an adaptive network-based fuzzy inference system (ANFIS) pattern recognition scheme is introduced. Monte Carlo computations were used to define the optimum handoff location for a mobile terminal. Also, a comparative performance evaluation of newly introduced asymmetric traditional and pattern recognition algorithms, designed for inter-technology handoff was performed for rate adaptive wireless networks, to enable the efficient classification of users mobility in rate adaptive networks.

III. System Model Design

A. Analytical Framework

During all initiation, Mobile users are more sensitive to call dropping than call blocking [12]. Hence, mobile network operators should focus at minimizing the call dropping probability. A call is dropped when a handoff failure occurs. The handoff failure probability, P_{HF} is given by [13]:

$$P_{HF} = \frac{\frac{1}{R!M!}\left[\frac{(\lambda_N+\alpha\lambda_H)^\gamma}{\mu_N+\mu_H}\right]^M\left[\frac{\alpha(\lambda_H^\gamma)}{\mu_H}\right]^R}{\sum_{S=0}^{M}\frac{1}{S!}\left[\frac{(\lambda_N+\alpha\lambda_H)^\gamma}{\mu_N+\mu_H}\right]^S+\sum_{S=(M+1)}^{C}\frac{1}{M!}\left[\frac{(\lambda_N+\alpha\lambda_H)^\gamma}{\mu_N+\mu_H}\right]^M\frac{1}{(S-M)!}\left[\frac{\alpha(\lambda_H^\gamma)}{\mu_H}\right]^{S-M}} \quad (1)$$

where, λ_N is the arrival rate for new calls; λ_H, the arrival rate for handoff calls; μ_N, the service rate for new calls; μ_H, the service rate for handoff calls; α, the mobility factor; γ, the signal strength; C, the number of channels; $S = C+1$, the channel allocation state; R, the reserved channels; and M, the shared channels.

We transform equation (1) into a form that is useful for this research by introducing the relevant parameters we are investigating. Assuming the channel holding time for new and handoff calls has an exponential distribution with mean, $\frac{1}{\mu}$, then, equation (1) can be rewritten as,

$$P_{HF} = \frac{\frac{1}{R!M!}\left[\frac{(\lambda_N+\alpha\lambda_H)^\gamma}{\mu}\right]^M\left[\frac{\alpha(\lambda_H^\gamma)}{\mu}\right]^R}{\sum_{S=0}^{M}\frac{1}{S!}\left[\frac{(\lambda_N+\alpha\lambda_H)^\gamma}{\mu}\right]^S+\sum_{S=(M+1)}^{C}\frac{1}{M!}\left[\frac{(\lambda_N+\alpha\lambda_H)^\gamma}{\mu}\right]^M\frac{1}{(S-M)!}\left[\frac{\alpha(\lambda_H^\gamma)}{\mu}\right]^{S-M}} \quad (2)$$

Now, let A_N and A_H, be the offered load and handoff traffic, respectively. From queuing theory, $A_N = \frac{\lambda_N}{\mu}$ and $A_H = \frac{\lambda_H}{\mu}$. Substituting these into equation (2) yields the analytical equation, thus,

$$P_{HF} = \frac{\frac{1}{R!M!}\left[(A_N+\alpha A_H)^\gamma\right]^M\left[\alpha(A_H^\gamma)\right]^R}{\sum_{S=0}^{M}\frac{1}{S!}\left[(A_N+\alpha A_H)^\gamma\right]^S+\sum_{S=(M+1)}^{C}\frac{1}{M!}\left[(A_N+\alpha A_H)^\gamma\right]^M\frac{1}{(S-M)!}\left[(\alpha A_H)^\gamma\right]^{S-M}} \quad (3)$$

where A_n, is defined as,

$$A_N = \frac{C_A}{(1-dropping\ probability)} = \frac{C_A}{1-P_d} \quad (4)$$

and,

C_A is the carried traffic

P_d is the dropping probability

The decision to handoff calls is guided by the guarantee that calls will be successfully handed over at the neighboring zone. To obtain the probability of successful handoff, equation (3) is subtracted from unity. The design of a framework to compare and evaluate the performance of handoff algorithms is indeed complex, as different networks have varying performance evaluation criteria. We have introduced an analytical solution to the handoff success rate problem, but, a full coverage of this solution will be discussed in an extended version of this paper. Henceforth, we proceed to discuss the neural network model in the next section.

B. Neural Network Model

Neural networks consist of a large class of varying architectures. In most cases, the concern is to approximate a static nonlinear mapping function, *f(x)* with a neural network. f_{NN}, where $x \in \Re^k$. In this paper, we adopt a neural network model, useful for function approximation – the Multilayer Perceptron (MLP). A MLP consists of an input layer, several hidden layers, and an output layer. The inputs ($x_k, k = 1,2,...,K$) to the neuron are multiplied by weights w_{ki} and accumulated with the constant bias term, θ_i. The resulting n_i is the input to the activation function *g*. A typical MLP architecture illustrating our system is shown in Figure 1.

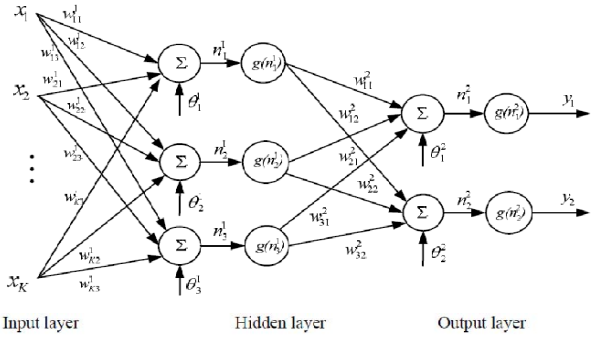
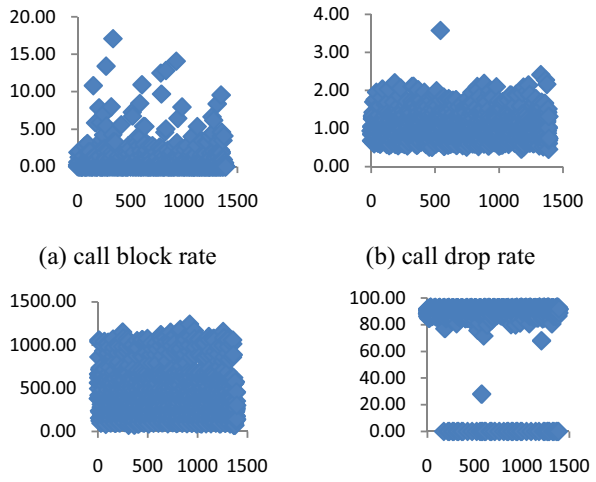


Figure 1. A multilayer perceptron network with one hidden layer [14]. Here the same activation function *g* is used in both layers. The superscript of *n*, θ, or w, refers to the first or second layer.

IV. Data Collection and Analysis

Data for this research were gathered from base stations servicing the south-east region of Nigeria, using a dashboard Motorola monitoring device. A total of forty (40) base stations belonging to one of the leading (CDMA) telecommunications carrier were covered. Figure 2 shows the distribution of data for the various factors/parameters considered in this paper, namely, *call block rate*, *call drop rate*, *carried traffic* and *signal strength* of mobile users from the base stations. From the Figure, we observed a high degree of inconsistency in the data patterns exhibited by the network.



(a) call block rate (b) call drop rate

(c) carried traffic (d) signal strength

Figure 2. Distribution of data obtained from a CDMA carrier operating in Nigeria

To investigate the effect of noise on the network, the Signal Interference to Noise Ratio (SINR) data were also obtained.

V. NN Model Simulation

A. Network Training

A total of 1391 sample datasets were obtained and trained by the NN model. The training data were distributed as follows: training – 70% (974 samples), test – 15% (208 samples), and validation – 15% (208 samples). Table I showss the first 15 inputs.

TABLE I. DATASET INPUTS

S/No.	Carried Traffic	Block_rate	Drop_rate	RSS(%)	SIR
1	380.70	0.00	0.94	88.22	-8.5
2	564.97	0.23	0.95	91.71	-10.5
3	496.43	0.08	1.05	90.61	-9.5
4	665.12	1.89	0.69	86.17	-9
5	622.68	0.55	0.86	89.00	-8
6	862.72	0.64	0.85	89.38	-9
7	1038.45	0.19	0.85	86.73	-8.5
8	1057.58	0.00	0.67	87.56	-10
9	657.82	0.05	1.07	89.25	-10
10	391.14	0.02	1.26	89.97	-10
11	420.75	0.06	1.33	91.29	-9.5
12	417.94	0.03	1.27	88.66	-9
13	239.80	0.00	1.30	90.43	-9
14	156.39	0.28	1.34	88.57	-10
15	1044.03	1.25	1.43	90.06	-10.5

During training, we adopted the Levenberg-Marquardt backpropagation algorithm (*trainlm*) – the fastest backpropagation algorithm in the MATLAB 2015b toolbox. The *trainlm* is a network training function that updates weight and bias values according to Levenberg-Marquardt optimization. The choice of this algorithm is based on the recommendation that it is a first-choice supervised algorithm, although it does require more memory than other algorithms.

B. Network Performance Analysis

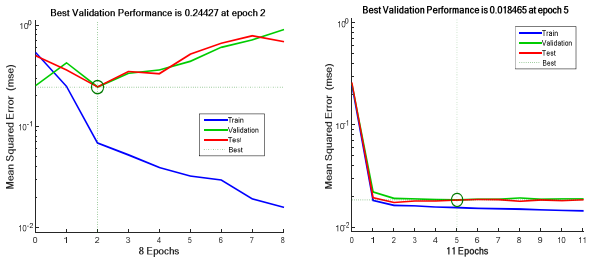
To achieve optimum results, our model was retrained several times with varying number of layers and neurons. We are interested in the following: the performance, the magnitude of performance gradient, and the number of validation checks. Table II catalogs selected trainings and their respective mean square error (MSE), obtained using varied number of layers and neurons.

TABLE II. RESULT OF SELECTED TRAININGS

S/No.	Data	No. of Layer	No. of Neuron	MSE
	with SINR data			
1	Small size	2	10	0.2443
2	Big size	2	10	0.0185
3	Small size	2	50	0.2961
4	Big size	2	50	0.0228
5	Small size	10	50	0.1406
6	Big size	10	50	0.2090
	with RSS data			
7	Normalized	10	50	0.0368
8	Not normalized	10	50	0.0357

With SINR data

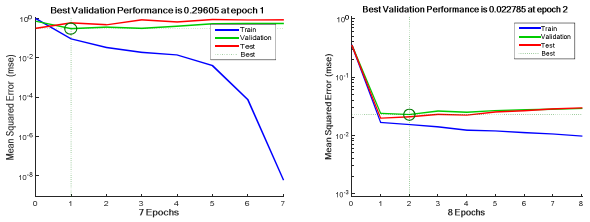
A training simulation was carried out with the following datasets: carried traffic, call block rate, call drop rate and SINR. In Figure 3 ((a) and (b)), the performance plots for two layers, 10 neurons NN for small and big datasets, respectively, are presented. We observed that for small data, the network performance was poorly predicted, but greatly improved for big data, as evidenced in the respective MSE values (i.e., 0.2443 and 0.0185).



(a) small dataset (b) big dataset

Figure 3. Performance plots for 2 layers, 10 neurons NN

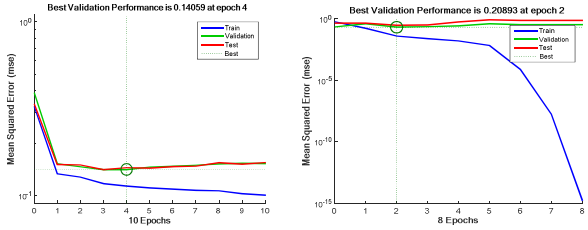
Similar performance plots were observed when the number of neurons was increased from 10 to 50. See Figure 4 ((a) and (b)).



(a) small dataset (b) big dataset

Figure 4. Performance plots for 2 layers, 50 neurons NN

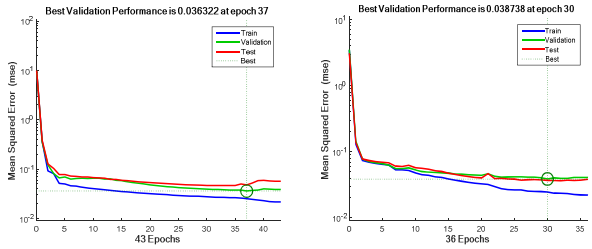
Increasing the number of layers did not however benefit the system, as the performance of the system, for both data sets worsen (see Figures 5 (a) and (b)).



(a) Small dataset (b) Big dataset

Figure 5. Performance plots for 10 layers, 50 neurons NN, for small and big datasets

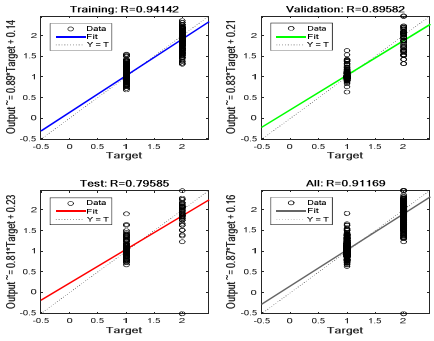
In Figure 6 ((a) and (b)), we observed that using the RSS dataset leveraged the prediction of the system, for both normalized and un-normalized input data – using big data. Hence the choice of RSS to the neighboring base stations, as a preferred figure of merit is here verified.



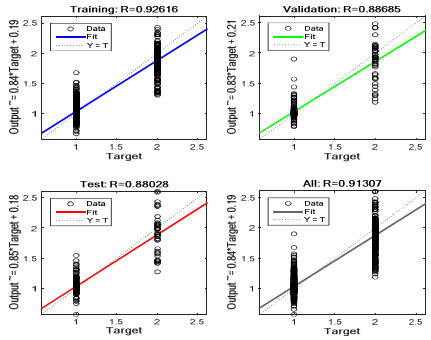
(a) normalized (b) un-normalized

Figure 6. Performance plots for 10 layers, 50 neurons NN, for normalized and un-normalized datasets

The regression plots for both normalized and un-normalized data are shown in Figures 6 and 7, respectively. We observed a near perfect correlation of training for the overall experiment (i.e., R>0.9).



(a) normalized



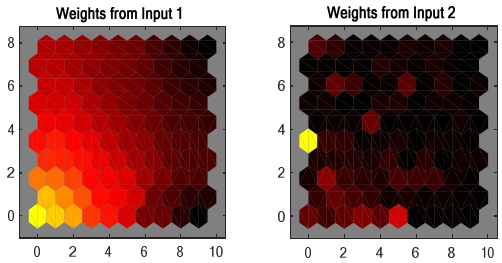
(b) un-normalized

Figure 7. Regression plots for normalized and un-normalized datasets

We proceed to scrutinize the existing system by visualizing each of the inputs using self organizing maps (SOMs).

C. Visualization of the Existing Network Topology using SOM

Self Organizing Feature Maps (SOFMs) learn to classify inputs according to their patterns in the input space. They differ from competitive layers in that neighboring neurons in that the self organizing map learn to recognize neighboring areas of the input space. Hence, SOMs learn not only the distribution (as do competitive layers), but also the topology of the inputs they are trained on. In Figure 8, the connection patterns of inputs 2 and 3 (i.e., call block rate and call drop rate) appear very similar, indicating that the inputs are highly correlated. But inputs 1 and 4 (i.e., carried traffic and signal strength) have dissimilar connections from the other two inputs, as the weights of input 4 are larger than those of input 1. The large deviations caused by these two inputs are not unconnected with the poor signal quality and very high traffic demands experienced by the existing network. Consequently, network operators should concentrate on the optimization of channels allocation and signal quality to help minimize the unabated calls blocking and dropping rates of their networks during handoffs.



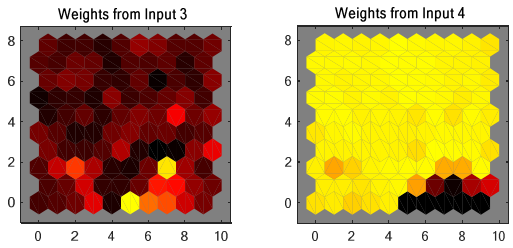


Figure 8. SOM weight planes for cellular network under study

Figure 9 reveals how many data points are associated with each neuron. Fairly evenly distributed data across the neurons is best. But, as observed in the Figure, data are concentrated a little more in the upper-right neurons, and the overall distribution is not fairly even. This implies that the data points associated with some of the nodes contribute less to the input pattern formation, and hence, the existing network certainly requires optimization of the clusters with widely deviated or uncorrelated weights, as a means to improve the uneven distribution of data in the network. The uneven distribution is not unconnected with the SOMs obtained in Figure 8, where the weights of inputs 2 and 3 contain small groups of tightly clustered data points, as opposed to the wider deviations introduced by inputs 1 and 4.

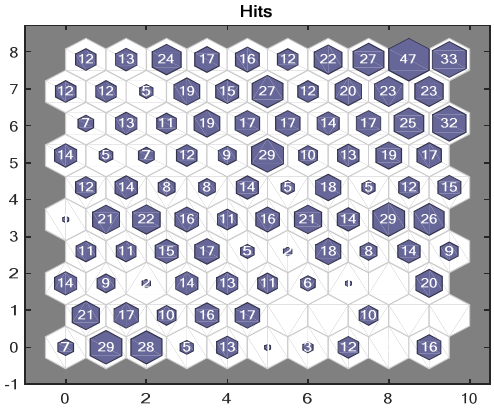


Figure 9. SOM sample hits for cellular network under study

VI. Conclusion and Future Works

Mobile Internet and multimedia services are becoming huge contributors of traffic in wireless communication. To support these services, the network infrastructure requires one or more points of attachment to enable it handle crucial operations such as mobility and handoffs. We have in this paper explored a more efficient approach to handoff decisions management by predicting in an intelligent way, several factors that contribute to the handoff decision itself. We identified these factors, collected empirical data from an existing telecommunications service provider, and simulated same using the proposed system. After subsequent learning of the empirical data, an optimized solution was obtained. Results of the optimized system were impressive and can be improved upon. A future direction of this research is a study of level of interactions between the various factors, to assist the prioritization of the decision variables and improve on the overall network performance.

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